

First Normal Form (1NF) — Relational Database Design

Definition: A relation is in **First Normal Form (1NF)** if and only if all attributes contain only **atomic (indivisible) values**, and there are no repeating groups or arrays stored within a single field.

1. Detailed Technical Explanation

Normalization is the systematic process of organizing relational database schemas to eliminate data redundancy and anomalies (Insertion, Deletion, and Update anomalies).

Key Rules of 1NF:

- 1. **Atomic Values:** Each column must store a single value (no multi-valued attributes like comma-separated phone numbers).
- 2. **Unique Column Names:** Each column in a table must have a unique attribute name.
- 3. **Unique Rows:** Each row must be uniquely identifiable (using a Primary Key).
- 4. **Order Indifference:** The order of rows and columns does not affect data meaning.

Conversion Example: Unnormalized Table to 1NF

Unnormalized Relation (UNF):

Student_ID	Student_Name	Phone_Numbers	Courses
101	Rahul Sharma	9876543210, 9123456789	DBMS, DSA, MPCA
102	Anita Verma	9988776655	DBMS, Python

Normalized 1NF Relation:

Student_ID	Student_Name	Phone_Number	Course
101	Rahul Sharma	9876543210	DBMS
101	Rahul Sharma	9876543210	DSA
101	Rahul Sharma	9876543210	MPCA
101	Rahul Sharma	9123456789	DBMS
101	Rahul Sharma	9123456789	DSA
101	Rahul Sharma	9123456789	MPCA
102	Anita Verma	9988776655	DBMS
102	Anita Verma	9988776655	Python

2. Core Concepts & Memory Keywords

- **Atomic Domains:** Fields storing single indivisible values.
 - **Repeating Groups:** Storing lists or multiple values in a single cell (violates 1NF).
 - **Primary Key:** A minimal set of attributes uniquely identifying each record in 1NF.
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3. Must-Write Points for Exams

- 1NF eliminates **multi-valued attributes** and **composite attributes** from relation instances.
 - In 1NF, every attribute value in a row must be a single scalar value from the domain.
 - 1NF does not eliminate all redundancy; partial and transitive dependencies can still cause anomalies.
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4. Quick Recall Flow

Unnormalized Table -> Remove Comma-Separated Values -> Ensure Atomic Cells -> 1NF Form Achieved